

Unit II - Applications of Differential Equations

chapter 2 :- Applications of Differential Eqs.

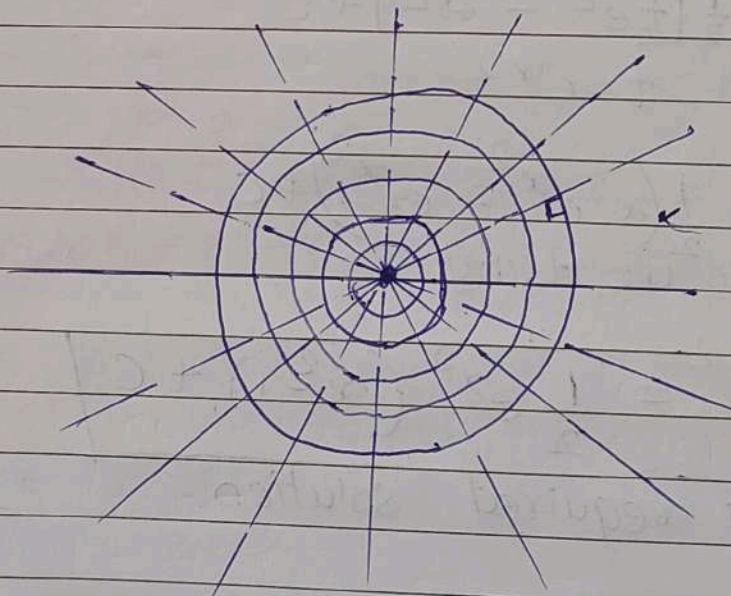
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① * Orthogonal Trajectories :-

* Trajectory :- A curve which cuts every member of a given family of curves according to some definite rule is called as trajectory of that family.

* Orthogonal Trajectory :- Two families of curves are said to be orthogonal trajectories of each other, if each curve in either family is orthogonal (that is perpendicular) to every curve in other family.

Ex) The family of line, $y = mx$ passing through origin and family of circles $x^2 + y^2 = a^2$ with centre $(0,0)$ and radius 'a' are orthogonal trajectories of each other.



* Steps to find orthogonal trajectory: - $f(x, y, c) = 0$

- ① Differentiate given function w.r.t. x to find $\frac{dy}{dx}$.
- ② Eliminate arbitrary constant if any, by using given equation
- ③ Arrange the given DE as $\frac{dy}{dx}$.
- ④ Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$
- ⑤ Solve the DE by Variable Separable form (V.S.)

* Polar Co-ordinates: $f(r, \theta, c) = 0$

- ① Differentiate w.r.t. θ to find $\frac{dr}{d\theta}$.
- ② Eliminate arbitrary constant if any.
- ③ Arrange DE as $\frac{dr}{d\theta}$
- ④ Replace $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$
- ⑤ Solve the DE by V.S. form.

Ex) Find the orthogonal trajectories of the family of straight lines, $y = mx$.

⇒ Given $y = mx$ —————→ ①
Differentiate w.r.t. x

$$\frac{dy}{dx} = \frac{d}{dx} (mx)$$

$$= m \frac{d}{dx} (x)$$

$$= m(1)$$

$$= m$$

$$\frac{dy}{dx} = m$$

put this value of m in eqn ①

$$\therefore y = \left(\frac{dy}{dx} \right) x$$

$$\therefore \frac{dy}{dx} = \frac{y}{x}$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ we get

$$-\frac{dx}{dy} = \frac{y}{x}$$

$$-x dx = y dy$$

$$-x dx - y dy = 0$$

$$x dx + y dy = 0$$

on integration

$$\int x dx + \int y dy = c$$

$$\frac{x^2}{2} + \frac{y^2}{2} = c$$

$$x^2 + y^2 = 2c$$

$$\boxed{x^2 + y^2 = c_1}$$

which is required orthogonal trajectories.

H.W. Que ① Find the orthogonal trajectories of the family of

(A) $y^2 = 4ax$

(B) $xy = c$

Que Find the orthogonal trajectories of the circles defined by $x = a \cos \theta$ which all pass through the origin and have their centres on the

Initial line. 'a' being the variable diameter.

$$\Rightarrow \text{Given } r = a \cos \theta \longrightarrow (1)$$

diff. w.r.t. θ

$$\frac{dr}{d\theta} = -a \sin \theta$$

but from (1) $a = \frac{r}{\cos \theta}$

$$\therefore \frac{dr}{d\theta} = -\frac{r}{\cos \theta} \times \sin \theta$$

$$\frac{dr}{d\theta} = -r \tan \theta$$

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta}$$

Replace $\frac{dr}{d\theta}$ by $-\frac{r^2 d\theta}{dr}$

$$\therefore -\frac{r^2 d\theta}{dr} = -r \tan \theta$$

$$\frac{d\theta}{\tan \theta} = \frac{dr}{r}$$

($\therefore$ divide by r^2 on b.s.)

on integration

$$\int \cot \theta d\theta = \int \frac{dr}{r} + \log c \quad \left(\frac{1}{\tan \theta} = \cot \theta \right)$$

$$\log(\sin \theta) = \log r + \log c$$

$$\cancel{\log(\sin \theta)} = \cancel{\log(r c)}$$

$$\sin \theta = r c$$

$$r = \frac{1}{c} \sin \theta$$

$$\therefore k = \frac{1}{c}$$

$$\boxed{r = k \sin \theta}$$

Que Find the orthogonal trajectory of $r^2 = a \sin 2\theta$

$$\Rightarrow r^2 = k \cos 2\theta$$

IMP ② Newton's Law of Cooling :-

* **Statement:-** The rate of temperature of the body is proportional to the difference between the temperature of body and that of surrounding medium.

i.e. $\frac{d\theta}{dt} \propto (\theta - \theta_0)$

$$\frac{d\theta}{dt} = -K(\theta - \theta_0)$$

where θ_0 = surrounding temperature

K = constant of proportionality

Ex) A body originally at 80°C cools to 60°C in 20 minutes, the temperature of air being 40°C . What will be the temperature of body after 40 minutes from the original?

⇒ Given;

step ① $\theta_1 = 80^\circ\text{C}$, $t_1 = 0 \text{ min}$

$\theta_2 = 60^\circ\text{C}$, $t_2 = 20 \text{ min}$

$\theta_3 = ?$, $t_3 = 40 \text{ min}$

and $\theta_0 = 40^\circ\text{C}$

($\because \theta_0$ is lowest temp., θ and t are increasing)

step ② By Newton's law of cooling

$$\frac{d\theta}{dt} = -K(\theta - \theta_0)$$

$$\frac{d\theta}{\theta - \theta_0} = -K dt$$

which is a variable separable form.

$$\int \frac{d\theta}{\theta - \theta_0} = - \int K dt + c$$

$$\ln(\theta - \theta_0) = -Kt + c \longrightarrow \text{①}$$

$$\therefore \ln(x) = \log_e(x).$$

step ③ find c

When $t=0$ then $\theta = 80^\circ\text{C}$, $\theta_0 = 40^\circ\text{C}$.

eqn ① becomes

$$\ln(80-40) = 0 + c$$

$$\boxed{c = \ln(40)}$$

put this value in eqn ① we get

$$\boxed{\ln(\theta - \theta_0) = -kt + \ln(40)} \longrightarrow \textcircled{2}$$

step ④ find ' $-k$ '.

When $t = 20 \text{ min}$, $\theta = 60^\circ\text{C}$.

$\therefore$ eqn ② becomes

$$\ln(60-40) = -k(20) + \ln(40)$$

$$\ln(20) - \ln(40) = -20k$$

$$\log A - \log B = \log\left(\frac{A}{B}\right)$$

$$\ln\left(\frac{20}{40}\right) = 20(-k)$$

$$\frac{1}{20} \ln\left(\frac{1}{2}\right) = -k$$

$$\boxed{k = \frac{1}{20} \ln\left(\frac{1}{2}\right)}$$

put this value of $-k$ in eqn ②

$$\boxed{\ln(\theta - \theta_0) = \frac{1}{20} \ln\left(\frac{1}{2}\right)t + \ln(40)} \longrightarrow \textcircled{3}$$

step ⑤ find θ_3 .

When $t = 40 \text{ min}$, eqn ③ becomes

$$\ln(\theta - 40) = \frac{1}{20} \ln\left(\frac{1}{2}\right) \times 40 + \ln(40)$$

$$\ln(\theta - 40) = 2 \ln\left(\frac{1}{2}\right) + \ln(40)$$

$$\ln(\theta - 40) = \ln\left(\frac{1}{2}\right)^2 + \ln(40) \quad \left(\log m^n = n \times \log m\right)$$

$$\cancel{\ln}(\theta - 40) = \cancel{\ln}\left(\frac{1}{2} \times 40\right)$$

$$\theta - 40 = 10$$

$$\boxed{\theta = 50^\circ}$$

Que A body at temperature 100°C is placed in a room whose temperature is 20°C and cools at 60°C in 5 minutes. Find the temperature after a further interval of 3 minutes?

⇒ Given,

$$\theta_1 = 100^{\circ}\text{C}, \quad t_1 = 0 \text{ min}$$

$$\theta_2 = 60^{\circ}\text{C}, \quad t_2 = 5 \text{ min}$$

$$\theta_3 = ? \quad t_3 = 5 + 3 = 8 \text{ min.}$$

$$\theta_0 = 20^{\circ}\text{C}.$$

By Newton's law of cooling

$$\frac{d\theta}{dt} = -k(\theta - \theta_0)$$

$$\frac{d\theta}{\theta - \theta_0} = -k dt \quad \text{v.s. form}$$

$$\theta - \theta_0$$

on integration

$$\int \frac{d\theta}{\theta - \theta_0} = - \int k dt + C$$

$$\boxed{\ln(\theta - \theta_0) = -kt + C} \longrightarrow \textcircled{1}$$

Now find C .

When $\theta = 100^{\circ}\text{C}$, $t = 0$ and $\theta_0 = 20^{\circ}\text{C}$.

eqn $\textcircled{1}$ becomes

$$\ln(100 - 20) = -k(0) + C$$

$$\boxed{C = \ln(80)}$$

put this value in eqn $\textcircled{1}$

$$\boxed{\ln(\theta - \theta_0) = -kt + \ln(80)} \longrightarrow \textcircled{2}$$

Now find $-k$.

When $t = 5 \text{ min}$ $\theta = 60^{\circ}\text{C}$.

eqn $\textcircled{2}$ becomes

$$\ln(60 - 20) = -k(5) + \ln(80)$$

$$\ln(40) = -5k + \ln(80)$$

$$\ln\left(\frac{40}{80}\right) = -5k$$

$$\therefore \left[-k = \frac{1}{5} \ln\left(\frac{1}{2}\right) \right]$$

put this value of $-k$ in eqn ②

$$\left[\ln(\theta - \theta_0) = \frac{1}{5} \ln\left(\frac{1}{2}\right) t + \ln(\theta_0) \right] \rightarrow \text{③}$$

Now find θ_3 .

when $t = 8$ min above eqn becomes

$$\ln(\theta - 20) = \frac{1}{5} \ln\left(\frac{1}{2}\right) \times 8 + \ln(80)$$

$$\ln(\theta - 20) = \frac{8}{5} \ln\left(\frac{1}{2}\right) + \ln(80)$$

$$\ln(\theta - 20) = \ln\left(\frac{1}{2}\right)^{8/5} + \ln(80)$$

$$\cancel{\ln(\theta - 20)} = \cancel{\ln\left(\left(\frac{1}{2}\right)^{8/5} \times 80\right)}$$

$$\theta = \frac{1}{(2^{8/5})} \times 80 + 20 \quad \text{--- calculator}$$

$$\boxed{\theta = 46.39^\circ\text{C}}$$

H.W. Que. A body of temp. 100°C is placed in a room whose temperature is 20°C and cools to 60°C in 5 minutes. What will be the temp. after 10 minutes?

$\Rightarrow \theta = 40^\circ\text{C}$

Que. A body of temp. 100°C is placed in a room whose temp. is 25°C and cools to 80°C in 10 minutes. Find the time when the temp. will be 60°C .

$\Rightarrow$ Given $\theta_1 = 100^\circ\text{C}$, $t_1 = 0$ min
 $\theta_2 = 80^\circ\text{C}$, $t_2 = 10$ min
 $\theta_3 = 60^\circ\text{C}$, $t_3 = ?$
 $\theta_0 = 25^\circ\text{C}$

By Newton's law of cooling,

$$\frac{d\theta}{dt} = -k(\theta - \theta_0)$$

$$\frac{d\theta}{\theta - \theta_0} = -k dt$$

(V.S. form.)

on integration

$$\int \frac{d\theta}{\theta - \theta_0} = - \int k dt + C$$

$$\boxed{\ln(\theta - \theta_0) = -kt + C} \longrightarrow \textcircled{1}$$

Now find C.

When $t = 0$, $\theta = 100^\circ\text{C}$, $\theta_0 = 25^\circ\text{C}$.
eqn $\textcircled{1}$ becomes

$$\ln(100 - 25) = -k(0) + C$$

$$\boxed{C = \ln(75)}$$

with this value eqn $\textcircled{1}$ becomes

$$\therefore \boxed{\ln(\theta - \theta_0) = -kt + \ln(75)} \longrightarrow \textcircled{2}$$

step $\textcircled{3}$ find $-k$.

When $t = 10 \text{ min}$, $\theta = 80^\circ\text{C}$.

eqn $\textcircled{2}$ becomes

$$\ln(80 - 25) = -k(10) + \ln(75)$$

$$\ln(55) - \ln(75) = -10k$$

$$\ln\left(\frac{55}{75}\right) = -10k$$

$$\boxed{-k = \frac{1}{10} \ln\left(\frac{11}{15}\right)}$$

Put this value in eqn $\textcircled{2}$

$$\boxed{\ln(\theta - \theta_0) = \frac{1}{10} \ln\left(\frac{11}{15}\right) t + \ln(75)} \longrightarrow \textcircled{3}$$

Now find t when $\theta = 60^\circ\text{C}$.

$$\therefore \ln(60 - 25) = \frac{1}{10} \ln\left(\frac{11}{15}\right) t + \ln(75)$$

on solving (calculator), $\boxed{t = 24.57 \text{ min}}$

Simplification: $\ln(35) - \ln(75) = \ln\left(\frac{11}{15}\right)^{\frac{1}{10}t}$

~~$\ln\left(\frac{35}{75}\right) = \ln\left(\frac{11}{15}\right)^{\frac{1}{10}t}$~~

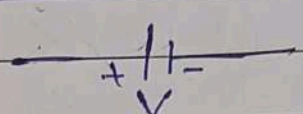
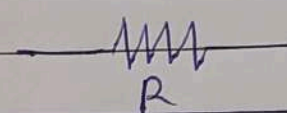
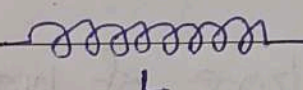
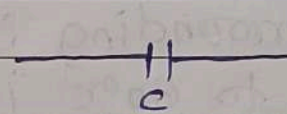
$\frac{7}{15} = \left(\frac{11}{15}\right)^{\frac{1}{10}t}$

$t = 24.57 \text{ min}$

que The temperature of water initially is 100°C and that of surrounding is 20°C . If the water cools down to 60°C in first 20 minutes, what will be the time required to fall temperature upto 30°C .

$\Rightarrow C = \ln(80), -k = \frac{1}{20} \ln\left(\frac{1}{2}\right), t = 60 \text{ min.}$

③ Simple Electrical Circuits :-

	Element	Symbol	Unit
1)	Voltage/ Battery		Volt
2)	Resistance		ohm (Ω)
3)	Inductance		Henry (H)
4)	Capacitance		Farad (F)

* The voltage drop across passive elements when current flows through an electrical circuits.

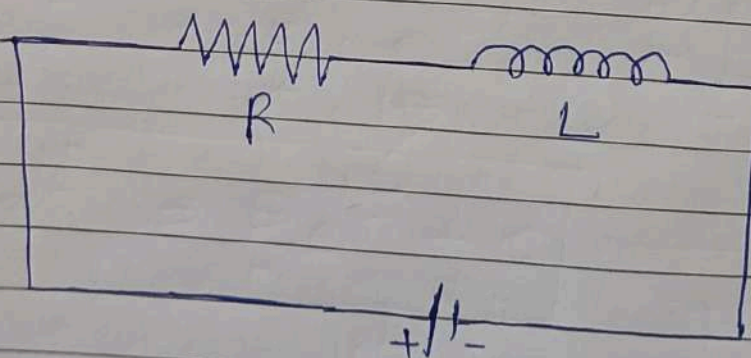
① Voltage drop across Resistance = Ri

② Voltage drop across Inductance = $L \frac{di}{dt}$

③ Voltage drop across Capacitance = $\frac{q}{C}$

④ Current = $i = \frac{dq}{dt}$, where q = charge.

* R-L Circuit :-

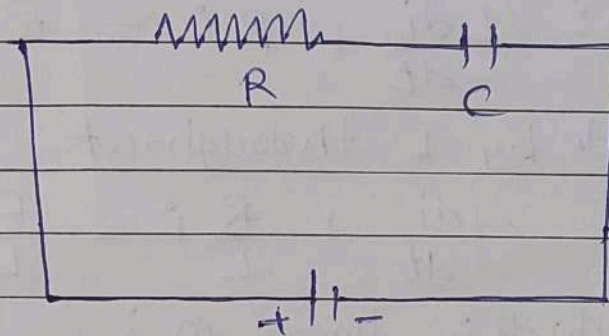


By Kirchhoff's law,
Voltage drop across R + Voltage drop across L = Applied voltage

$$Ri + L \frac{di}{dt} = E$$

This represents D.E. for R-L circuit.

* R.C. Circuit:-



By Kirchhoff's law,

$$\left[\text{Voltage drop across } R \right] + \left[\text{Voltage drop across } C \right] = \left[\text{Applied Voltage} \right]$$

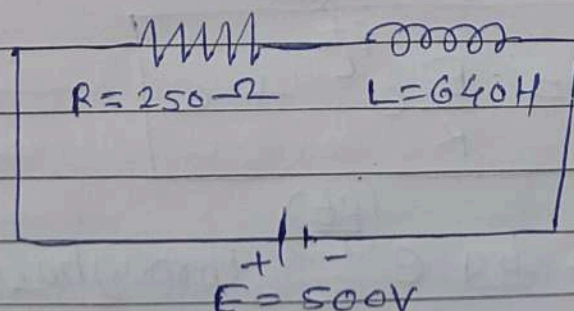
$$Ri + \frac{q}{C} = E$$

$$\text{but } i = \frac{dq}{dt}$$

$$R \frac{dq}{dt} + \frac{q}{C} = E$$

Que: A resistance of 250 ohms and an inductance of 640 Henry are connected in series with a battery of 500 volts. Find the current in the circuit if $i=0$ at $t=0$.

⇒



By Kirchhoff's law for electrical circuits,

$$\left[\text{Voltage drop across } R \right] + \left[\text{voltage drop across } L \right] = \left[\text{Applied Voltage} \right]$$

$$Ri + L \frac{di}{dt} = E$$

$$\therefore L \frac{di}{dt} + Ri = E$$

divide by L throughout

$$\frac{di}{dt} + \frac{R}{L} i = \frac{E}{L}$$

which is linear in i.

$$\therefore \text{where } P = \frac{R}{L} \text{ and } Q = \frac{E}{L}$$

$$I.F. = e^{\int P dx}$$

$$I.F. = e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}}$$

The general solution is given by

$$i (I.F.) = \int Q (I.F.) dt + C$$

$$i e^{\frac{Rt}{L}} = \int \frac{E}{L} \times e^{\frac{Rt}{L}} dt + C$$

$$i e^{\frac{Rt}{L}} = \frac{E}{L} \int e^{\frac{Rt}{L}} dt + C$$

$$\therefore i e^{\left(\frac{Rt}{L}\right)} = \frac{E}{L} \cdot \frac{e^{\left(\frac{Rt}{L}\right)}}{\left(\frac{R}{L}\right)} + C$$

$$\boxed{i e^{\left(\frac{Rt}{L}\right)} = \frac{E}{R} e^{\frac{Rt}{L}} + C} \longrightarrow (1)$$

Divide by $e^{\left(\frac{Rt}{L}\right)}$ throughout

$$\frac{i e^{\frac{Rt}{L}}}{e^{\frac{Rt}{L}}} = \frac{E e^{\frac{Rt}{L}}}{R e^{\frac{Rt}{L}}} + \frac{C}{e^{\frac{Rt}{L}}}$$

$$\boxed{i = \frac{E}{R} + C e^{-\frac{Rt}{L}}} \longrightarrow (2)$$

By initial conditions,

when $t=0 \Rightarrow i=0$

$$0 = \frac{E}{R} + C e^0$$

$$-\frac{E}{R} = C (1) \quad (e^0=1)$$

$$\boxed{C = -\frac{E}{R}}$$

put this value of C in eqn (2)

$$i = \frac{E}{R} - \frac{E}{R} e^{-\frac{Rt}{L}}$$

$$\boxed{i = \frac{E}{R} \left(1 - e^{-\frac{Rt}{L}}\right)}$$

this represent current as a function of t .

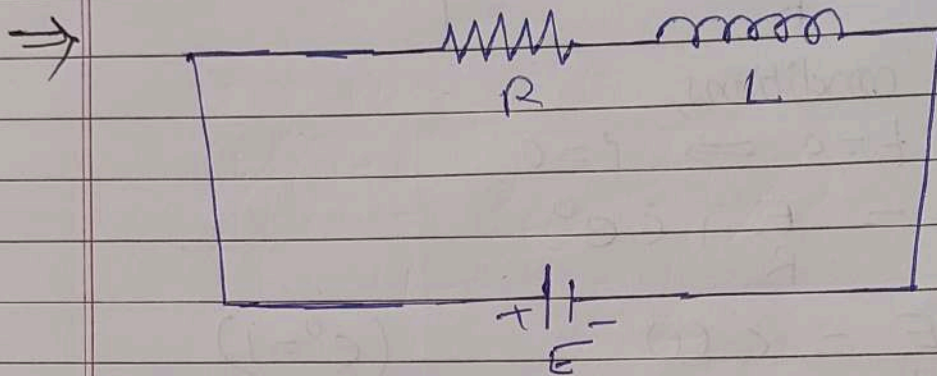
but here $R = 250 \Omega$, $L = 640 H$, and $E = 500V$

$$i = \frac{500}{250} \left(1 - e^{-\frac{250}{640} t}\right)$$

$$\boxed{i = 2 \left(1 - e^{-\frac{25}{40} t}\right)}$$

Que: A constant electromotive force E volts is applied to a circuit containing a constant resistance R ohms in a series and a constant inductance L henries. IF the initial

current is zero, show that the current builds up to half its theoretical maximum in $\frac{L}{R} \ln(2)$ seconds.



By Kirchhoff's law,

$$\left[\text{Voltage drop across } R \right] + \left[\text{Voltage drop across } L \right] = \left[\text{Applied Voltage} \right]$$

$$Ri + L \frac{di}{dt} = E$$

$$\frac{L di}{dt} + Ri = E$$

Divide by L throughout

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

which is linear in i

where $P = \frac{R}{L}$ and $Q = \frac{E}{L}$

$$I.F. = e^{\int P dx} = e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}} = e^{\frac{Rt}{L}}$$

and the general solution is given by

$$i(I.F.) = \int Q(I.F.) dt + c$$

$$i e^{\frac{Rt}{L}} = \int \frac{E}{L} e^{\frac{Rt}{L}} dt + c$$

$$i e^{\frac{Rt}{L}} = \frac{E}{\cancel{L}} \cdot \frac{e^{\frac{Rt}{L}}}{\cancel{\left(\frac{R}{L}\right)}} + c$$

$$\therefore i e^{\frac{Rt}{L}} = \frac{E e^{\frac{Rt}{L}}}{R} + c \quad \left(\because \int e^{at} dt = \frac{e^{at}}{a} \right)$$

Divide by $e^{\frac{Rt}{L}}$ throughout

$$i = \frac{E}{R} + c e^{-\frac{Rt}{L}} \longrightarrow \textcircled{1}$$

Apply initial conditions when $t=0 \Rightarrow i=0$

$$0 = \frac{E}{R} + c e^0$$

$$\boxed{c = -\frac{E}{R}}$$

eqn $\textcircled{1}$ becomes

$$i = \frac{E}{R} - \frac{E}{R} e^{-\frac{Rt}{L}}$$

$$\boxed{i = \frac{E}{R} \left(1 - e^{-\frac{Rt}{L}}\right)} \longrightarrow \textcircled{A}$$

This represents current as a function of time.

Now maximum current is obtained when time is maximum

i.e. when $t = \infty \Rightarrow i = i_{\max}$

$$\therefore i_{\max} = \frac{E}{R} (1 - e^{-\infty})$$

$$i_{\max} = \frac{E}{R} (1 - 0) \quad (\because e^{-\infty} = 0)$$

$$\boxed{i_{\max} = \frac{E}{R}} \longrightarrow \textcircled{B}$$

But $i = 50\%$ $i_{\max}$

$$i = \frac{50}{100} i_{\max}$$

$$i = \frac{1}{2} i_{\max}$$

from (A) and (B)

$$\frac{E}{R} (1 - e^{-\frac{Rt}{L}}) = \frac{1}{2} \left(\frac{E}{R} \right)$$

$$1 - e^{-\frac{Rt}{L}} = \frac{1}{2}$$

$$1 - \frac{1}{2} = e^{-\frac{Rt}{L}}$$

$$\frac{1}{2} = e^{-\frac{Rt}{L}}$$

Taking log on both sides

$$\ln\left(\frac{1}{2}\right) = \ln(e)^{-\frac{Rt}{L}}$$

$$\ln\left(\frac{1}{2}\right) = -\frac{Rt}{L} \quad (\because \ln(e) = 1)$$

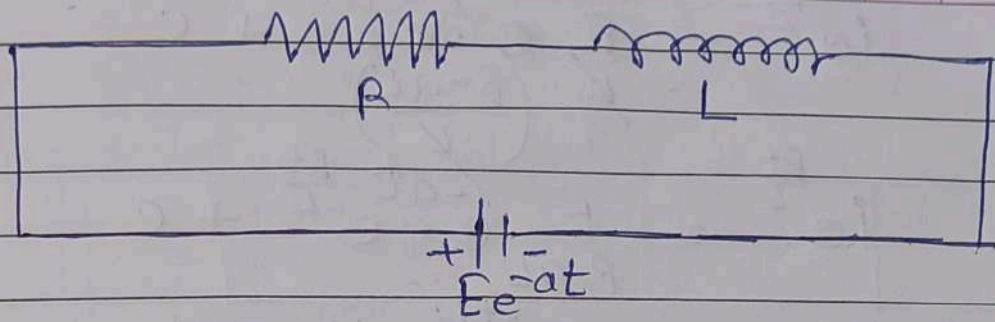
$$-\frac{R}{L} \ln\left(\frac{1}{2}\right) = t$$

$$t = \frac{L}{R} \ln(2) \quad \log\left(\frac{A}{B}\right) = -\log\left(\frac{B}{A}\right)$$

$$\therefore t = \frac{L}{R} \ln(2)$$

Que H.W. A voltage Ee^{-at} is applied at $t=0$ to a circuit containing inductance L , and Resistance R . Show that current at any time t is given by,

$$i = \frac{E}{(R - aL)} \left[e^{-at} - e^{-\frac{Rt}{L}} \right]$$



By Kirchoff's law,

$$\left[\text{voltage drop across } R \right] + \left[\text{voltage drop across } L \right] = \left[\text{Applied Voltage} \right]$$

$$Ri + L \frac{di}{dt} = Ee^{-at}$$

$$L \frac{di}{dt} + Ri = Ee^{-at}$$

divide by L throughout,

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}e^{-at}$$

which is linear in i

where $p = \frac{R}{L}$ and $Q = \frac{E}{L}e^{-at}$

$$I.F. = e^{\int p dx} = e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}}$$

The general solution is given by

$$i(I.F.) = \int Q(I.F.) dt + C$$

$$ie^{\frac{Rt}{L}} = \int \frac{E}{L} e^{-at} \cdot e^{\frac{Rt}{L}} dt + C$$

$$ie^{\left(\frac{Rt}{L}\right)} = \frac{E}{L} \int e^{\left(\frac{R}{L} - a\right)t} dt + C \quad \left(\begin{matrix} m & n & m+n \\ a \times a^m = a^{m+n} \end{matrix} \right)$$

$$ie^{\left(\frac{Rt}{L}\right)} = \frac{E}{L} \cdot \frac{e^{\left(\frac{R}{L} - a\right)t}}{\left(\frac{R}{L} - a\right)} + C$$

$$i e^{\frac{Rt}{L}} = \frac{E}{\cancel{R-aL}} \frac{e^{\frac{Rt}{L}} e^{-at}}{\cancel{e^{\frac{Rt}{L}}}} + c$$

$$i e^{\frac{Rt}{L}} = \frac{E}{R-aL} e^{-at} e^{\frac{Rt}{L}} + c$$

divide by $e^{\frac{Rt}{L}}$ throughout

$$i = \frac{E}{R-aL} e^{-at} + c e^{-\frac{Rt}{L}} \quad \text{--- (1)}$$

Apply initial conditions to find c
when $t=0$, $i=0$

$$0 = \frac{E}{R-aL} e^0 + c e^0$$

$$c = -\frac{E}{(R-aL)}$$

eqn (1) becomes

$$i = \frac{E}{R-aL} e^{-at} - \frac{E}{R} e^{-\frac{Rt}{L}}$$

$$i = \frac{E}{R-aL} \left[e^{-at} - e^{-\frac{Rt}{L}} \right]$$

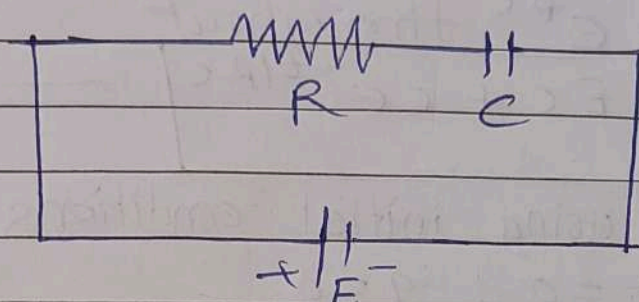
* Examples on R-C Circuit:-

Que

A circuit consists of resistance R ohms and condenser of C farads connected to a constant e.m.f. E volts. If q is the voltage of condenser at time t after closing the circuit, show that

$$\frac{q}{C} = E \left(1 - e^{-\frac{t}{RC}} \right)$$

⇒



By Kirchhoff's voltage law,

$$\left[\text{voltage drop across } R \right] + \left[\text{voltage drop across } C \right] = \left[\text{Applied voltage} \right]$$

$$Ri + \frac{q}{C} = E$$

$$\text{but } i = \frac{dq}{dt}$$

$$\therefore R \frac{dq}{dt} + \frac{q}{C} = E$$

divide by R throughout

$$\frac{dq}{dt} + \frac{1}{RC} q = \frac{E}{R}$$

which is linear in q .

$$\text{where } p = \frac{1}{RC} \text{ and } Q = \frac{E}{R}$$

$$I.F. = e^{\int p dx} = e^{\int \frac{1}{RC} dt} = e^{\frac{t}{RC}}$$

The general solution is given by

$$q(I.F.) = \int Q(I.F.) dt + C$$

$$q(e^{\frac{t}{RC}}) = \int \frac{E}{R} e^{\frac{t}{RC}} dt + C$$

$$q e^{\frac{t}{RC}} = \frac{E}{R} \frac{e^{\frac{t}{RC}}}{\left(\frac{1}{RC}\right)} + C \quad \left(\because \int e^{at} dt = \frac{e^{at}}{a} \right)$$

$$q e^{\frac{t}{RC}} = E c e^{\frac{t}{RC}} + C$$

divide by $e^{\frac{t}{RC}}$ throughout

$$\boxed{q = E c + c e^{-\frac{t}{RC}}} \longrightarrow \textcircled{1}$$

find c by using initial conditions
when $t=0$, $q=0$

$$\therefore 0 = E c + c e^0$$

$$\boxed{c = -E c}$$

eqn ① becomes

$$q = E c - E c e^{-\frac{t}{RC}}$$

$$q = E c (1 - e^{-\frac{t}{RC}})$$

divide by c on both sides

$$\frac{q}{c} = E (1 - e^{-\frac{t}{RC}})$$

but $\frac{q}{c} = \text{voltage}$

$$\therefore \boxed{\text{Voltage} = E (1 - e^{-\frac{t}{RC}})}$$

Que) The charge q on the plate of condenser of capacity ' c ' charged through a resistance R by a steady state voltage V satisfies the

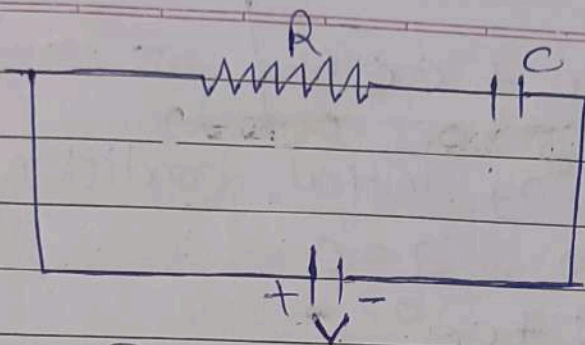
$$\text{D.E.} \quad R \frac{dq}{dt} + \frac{q}{c} = V$$

If $q=0$ at $t=0$, show that

$$q = cV [1 - e^{-\frac{t}{RC}}]$$

Also find the current flowing into the plate.

$\Rightarrow$ The given D.E. is $R \frac{dq}{dt} + \frac{q}{c} = V$



$$\left[\text{Voltage drop across } R \right] + \left[\text{Voltage drop across } C \right] = \left[\text{Applied Voltage} \right]$$

$$Ri + \frac{q}{C} = V$$

$$\text{but } i = \frac{dq}{dt}$$

$$\therefore R \frac{dq}{dt} + \frac{q}{C} = V$$

Divide by R throughout

$$\frac{dq}{dt} = \frac{q}{RC} - \frac{V}{R}$$

which is linear in q , where $P = \frac{1}{RC}$ and $Q = \frac{V}{R}$

$$\text{I.F.} = e^{\int P dt} = e^{\int \frac{1}{RC} dt} = e^{\frac{t}{RC}}$$

The general solution is given by

$$q(\text{I.F.}) = \int Q(\text{I.F.}) dt + C$$

$$q e^{\frac{t}{RC}} = \int \frac{V}{R} e^{\frac{t}{RC}} dt + C$$

$$q e^{\frac{t}{RC}} = \frac{V}{R} \frac{e^{\left(\frac{t}{RC}\right)}}{\left(\frac{1}{RC}\right)} + C \quad \left(\because \int e^{at} dt = \frac{e^{at}}{a} \right)$$

$$q e^{\frac{t}{RC}} = V C e^{\frac{t}{RC}} + C$$

divide by $e^{\frac{t}{RC}}$ throughout

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$$q = VC + Ce^{-\frac{t}{RC}} \quad \text{---} \rightarrow \textcircled{h}$$

find c by using initial conditions

when $t=0$, $q=0$.

$$0 = VC + ce^0$$

$$\boxed{c = -VC}$$

eqn ① becomes

$$q = VC - VCe^{-t/RC}$$

$$\boxed{q = VC(1 - e^{-t/RC})}$$

Now to find current, diff. above eqn w.r.t. t .

$$\frac{dq}{dt} = \frac{d}{dt} VC(1 - e^{-t/RC})$$

$$i = VC(0 - e^{-t/RC} \times \left(-\frac{1}{RC}\right))$$

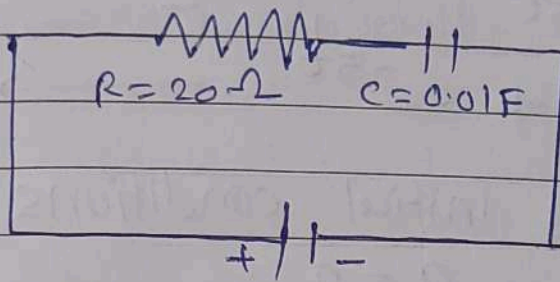
$$i = VC \cancel{e^{-t/RC}} \times \frac{1}{RC}$$

$$\boxed{i = \frac{V}{R} e^{-t/RC}}$$

Which is current flowing through plate at time t .

Que An em.f. of $200e^{-5t}$ is applied to a series circuit containing of 20 ohm resistor and 0.01F capacitor. Find the charge and current at any time assuming that there is no initial charge on the capacitor.

$\Rightarrow$



By kirchoff's law .

$$[\text{Voltage drop across } R] + [\text{Voltage drop across } C] = [\text{Applied voltage}]$$

$$Ri + \frac{q}{C} = 200e^{-5t}$$

Given $R = 20\Omega$, $C = 0.01F$, Also $i = \frac{dq}{dt}$

$$\therefore 20 \frac{dq}{dt} + \frac{q}{0.01} = 200e^{-5t}$$

Divide by 20 throughout

$$\frac{dq}{dt} + \frac{q}{20 \times 0.01} = \frac{200e^{-5t}}{20}$$

$$\frac{dq}{dt} + 5q = 10e^{-5t} \quad \left(\because \frac{1}{0.2} = \frac{10}{2} = 5 \right)$$

which is LDE in q .

where $p = 5$ and $Q = 10e^{-5t}$.

$$\therefore \text{I.F.} = e^{\int p dx} = e^{\int 5 dt} = e^{5t}$$

The general solution is

$$q \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dt + C$$

$$qe^{5t} = \int 10e^{-5t} \cdot e^{5t} dt + C$$

$$qe^{5t} = 10t + C$$

Dividing by e^{5t} throughout
 $q = 10te^{-5t} + ce^{-5t} \longrightarrow \textcircled{1}$

find c by using initial conditions
 when $t = 0$, $q = 0$

$$\therefore 0 = 10(0)e^0 + ce^0$$

$$\therefore 0 = 0 + c(1)$$

$$\boxed{c = 0}$$

With this value eqⁿ $\textcircled{1}$ becomes

$$q = 10te^{-5t} + 0e^{-5t}$$

$$\boxed{q = 10te^{-5t}}$$

Which is the charge at time t .
 to find current diff. w.r.t.

$$\frac{dq}{dt} = 10 \frac{d}{dt}(te^{-5t})$$

$$i = 10 \left[t \frac{d}{dt}(e^{-5t}) + e^{-5t} \frac{d}{dt}(t) \right]$$

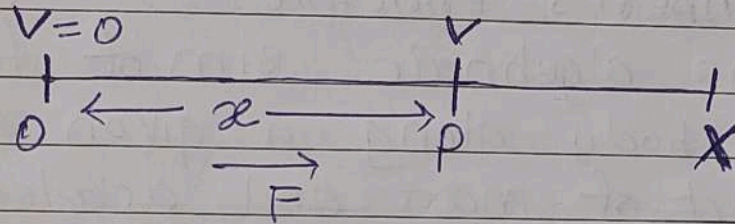
$$i = 10 \left[te^{-5t}(-5) + e^{-5t}(1) \right]$$

$$i = 10 [1 - 5t] e^{-5t}$$

$$\boxed{i = 10e^{-5t}(1 - 5t)}$$

* **Rectilinear Motion**:- Rectilinear motion is a motion of a body along a straight line.

Let a body of mass m start moving from a fixed point 'o' along x-axis. Let P be the position of particle at time t and $op = x$ then



(a) **The position (x)**:- The position of a particle is defined by the distance x between the particle and a fixed origin 'o' on the straight line.

(b) **The Velocity (v)**:- The velocity of a particle at any instant is defined by the rate of change of its position with respect to time. i.e. $v = \frac{dx}{dt}$

(c) **Acceleration**:- It is defined as the rate of change of particle's velocity.

$$a = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d^2x}{dt^2}$$

* **Displacement**:- The displacement of a particle during a certain time interval is defined as the change of its position.

- * Newton's second law of motion:-
 The rate of change of momentum of a body is proportional to the resultant force acting on a body.
 $\therefore \text{Net Force} = \text{mass} \times \text{acceleration}.$

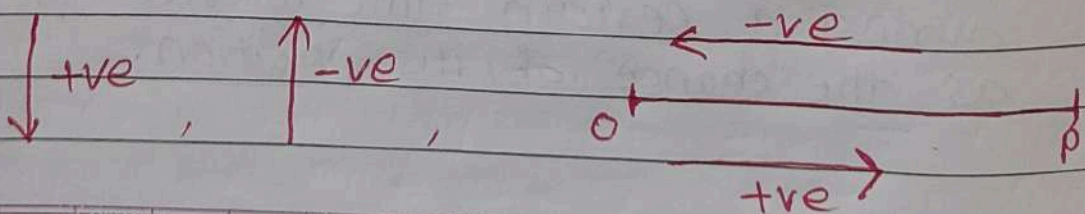
- * D'Alembert's Principle:-
 The algebraic sum of forces acting on a body along a given direction is the product of mass and acceleration.
 $\therefore \text{Net Force} = \text{mass} \times \text{acceleration}.$

* Relations:-

- ① $a = \frac{dv}{dt} \Rightarrow$ Relation betⁿ velocity and time (t) (v)
- ② $a = v \frac{dv}{dx} \Rightarrow$ Relation betⁿ velocity and distance (x) (v)
- ③ $a = \frac{d^2x}{dt^2} \Rightarrow$ Relation betⁿ distance and time (t) (x)

- * Note:- ① Downward forces are taken as positive and upward forces are taken as negative.

- ② Horizontal forces away from origin are taken as positive and towards the origin are taken as negative.



Ques) A body of mass m , falling from rest is subject to the force of gravity and an air resistance proportional to the square of velocity (kV^2). If it falls through a distance ' x ' and possesses a velocity V at that instant. Prove that

$$\frac{2Kx}{m} = \log\left(\frac{a^2}{a^2 - V^2}\right) \quad \text{where } mg = Ka^2.$$

⇒ The forces acting on body are:

- (i) Force of gravity (acting downwards) = mg
- (ii) Air resistance = $-kV^2$

$$\therefore \text{Net Force} = mg + (-kV^2)$$

$$\text{Net Force} = mg - kV^2$$

By D'Alembert's principle, the eqn of motion is
mass $\times$ acceleration = Net Force.

$$m \times \boxed{\frac{V dV}{dx}} = mg - kV^2$$

(In this problem we need relation betⁿ velocity and distance so relation (2) is used)

$$\text{but } mg = Ka^2$$

$$\therefore mV \frac{dV}{dx} = Ka^2 - kV^2$$

$$mV \frac{dV}{dx} = k(a^2 - V^2)$$

$$\therefore \frac{V}{a^2 - V^2} dV = \frac{k}{m} dx \quad (\text{V.S. form})$$

on integration.

$$\int \frac{V}{a^2 - V^2} dV = \int \frac{k}{m} dx + C$$

$$-\frac{1}{2} \int \frac{-2v}{a^2 - v^2} dv = \frac{k}{m} \int dx + C$$

$$\boxed{-\frac{1}{2} \log(a^2 - v^2) = \frac{k}{m} x + C} \longrightarrow \textcircled{1}$$

find C by initial conditions,

when $x=0$, $v=0$

$\therefore$ eqn $\textcircled{1}$ becomes

$$-\frac{1}{2} \log(a^2 - 0^2) = \frac{k}{m} (0) + C$$

$$\boxed{C = -\frac{1}{2} \log(a^2)}$$

put this value of C in eqn $\textcircled{1}$

$$\therefore -\frac{1}{2} \log(a^2 - v^2) = \frac{k}{m} x - \frac{1}{2} \log(a^2)$$

$$-\frac{1}{2} \log(a^2) - \frac{1}{2} \log(a^2 - v^2) = \frac{k}{m} x$$

$$\frac{1}{2} [\log a^2 - \log(a^2 - v^2)] = \frac{kx}{m}$$

$$\log A - \log B = \log \left(\frac{A}{B} \right)$$

multiply by 2 throughout

$$\boxed{\log \left(\frac{a^2}{a^2 - v^2} \right) = \frac{2kx}{m}} \therefore$$

Hence proved

Que) A particle of mass m is projected upwards with velocity v . Assuming the air resistance k times it's velocity, Write the eqn of motion and show it will reach maximum height in time

$$\frac{m}{k} \log \left(1 + \frac{kV}{mg} \right)$$

⇒ The forces acting on body are:

- i) Force of gravity = $-mg$
(particle projected upward, therefore force of gravity is -ve)
- ii) Air resistance = $-kV$

The Net force acting on body = $-mg - kV$

By D'Alembert's principle,

mass × Acceleration = Net Force

$$m \frac{dv}{dt} = -mg - kV$$

By D'Alembert's principle,

Mass × Acceleration = Net Force

$$m \frac{dv}{dt} = -mg - kV$$

$$m \frac{dv}{dt} = -(mg + kV)$$

$$\frac{m dv}{mg + kV} = -dt \quad (\text{v.s. form})$$

on integration

$$m \int \frac{1}{mg + kV} dv = - \int dt + C$$

$$\left(\int \frac{f'(x)}{f(x)} dx = \log(f(x)) + C \right) \quad m \int \frac{k}{mg + kV} dv = -t + C$$

$$\boxed{\frac{m}{k} \log(mg + kV) = -t + C} \rightarrow (1)$$

Now find C by initial conditions,

when $t=0$, $V=V$

eqn (1) becomes

$$\frac{m}{k} \log(mg + kV) = 0 + C$$

$$\therefore \boxed{C = \frac{m}{k} \log(mg + kv_0)}$$

with this value eqⁿ (1) becomes.

$$\frac{m}{k} \log(mg + kv) = -t + \frac{m}{k} \log(mg + kv_0)$$

$$\therefore t = \frac{m}{k} \log(mg + kv_0) - \frac{m}{k} \log(mg + kv)$$

$$t = \frac{m}{k} [\log(mg + kv_0) - \log(mg + kv)]$$

$$\boxed{t = \frac{m}{k} \left[\log \left(\frac{mg + kv_0}{mg + kv} \right) \right]} \rightarrow (2)$$

This represents time 't' at any instant throughout the motion.

∴ But as soon as the particle attains the maximum height, velocity becomes zero.

∴ let $t = t_1$ (at maximum height), $v = 0$

$$t_1 = \frac{m}{k} \log \left(\frac{mg + kv_0}{mg + 0} \right)$$

$$\boxed{t_1 = \frac{m}{k} \log \left(1 + \frac{kv_0}{mg} \right)} \quad \text{Hence proved.}$$

Que: A bullet is fired into a sand tank, its retardation is proportional to the square root of its velocity i.e. $(k\sqrt{v})$. How long it will take to come rest if it enters the sand tank with an initial velocity v_0 .

⇒ Acceleration = - retardation.

but given retardation = $k\sqrt{v}$.

$$\therefore \text{Accel}^n = -k\sqrt{v}$$

$$\therefore \frac{dv}{dt} = -k\sqrt{v}$$

$$\therefore \frac{dv}{\sqrt{v}} = -k dt \quad (V.s. form)$$

on integration

$$\int \frac{dv}{\sqrt{v}} = \int -k dt + C$$

$$\therefore \int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + C$$

$$\therefore \boxed{2\sqrt{v} = -kt + C} \longrightarrow \textcircled{1}$$

By initial conditions when $t=0$, $v=v_0$

$$\therefore 2\sqrt{v_0} = -k(0) + C$$

$$\boxed{C = 2\sqrt{v_0}}$$

eqn $\textcircled{1}$ becomes

$$\boxed{2\sqrt{v} = -kt + 2\sqrt{v_0}} \longrightarrow \textcircled{2}$$

Now we have to find out, how long it will take to come to rest.

when $v=0$ find t .

$$\therefore 2\sqrt{0} = -kt + 2\sqrt{v_0}$$

$$0 = -kt + 2\sqrt{v_0}$$

$$kt = 2\sqrt{v_0}$$

$$\boxed{t = \frac{2\sqrt{v_0}}{k}}$$

This is the required time.

Que H.W. A particle is moving in a straight line with an acceleration $k\left(x + \frac{a^4}{x^3}\right)$ directed

towards origin. If it starts from rest at a distance 'a' from the origin, prove that it will arrive at origin at the end of time $\frac{\pi}{4\sqrt{k}}$.

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{a} \tanh^{-1}\left(\frac{x}{a}\right) + C$$

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Que: The distance x descended by a parachute satisfies the D.E. $v \frac{dv}{dx} = g \left(1 - \frac{v^2}{k^2}\right)$ where v is velocity, k, g are constants. If $v = 0$ and $x = 0$ at time $t = 0$, show that $x = \frac{k^2}{g} \log(\cosh(\frac{gt}{k}))$

⇒ Given D.E.

$$v \frac{dv}{dx} = g \left(1 - \frac{v^2}{k^2}\right)$$

but- $v \frac{dv}{dx} = \frac{dv}{dt}$

$$\frac{dv}{dt} = g \left(1 - \frac{v^2}{k^2}\right)$$

$$\frac{dv}{dt} = g \left(\frac{k^2 - v^2}{k^2}\right)$$

$$\frac{dv}{k^2 - v^2} = \frac{g}{k^2} dt \quad \dots \text{v.s. form}$$

on integration

$$\int \frac{dv}{k^2 - v^2} = \frac{g}{k^2} \int dt + C$$

$$\boxed{\frac{1}{k} \tanh^{-1}\left(\frac{v}{k}\right) = \frac{gt}{k^2} + C} \rightarrow (1)$$

By using initial conditions,
when $t = 0$, $v = 0$

$$\frac{1}{k} \tanh^{-1}(0) = \frac{g}{k^2}(0) + C$$

$$0 = 0 + C$$

$$\boxed{C = 0}$$

eaⁿ (1) becomes

$$\frac{1}{k} \tanh^{-1}\left(\frac{v}{k}\right) = \frac{gt}{k^2} + 0$$

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$$\tanh^{-1}\left(\frac{v}{k}\right) = \cancel{k} \cdot \frac{g}{\cancel{k^2}} t$$

$$\therefore \tanh^{-1}\left(\frac{v}{k}\right) = \frac{g}{k} t$$

$$\frac{v}{k} = \tanh\left(\frac{gt}{k}\right)$$

$$\text{but } v = \frac{dx}{dt}$$

$$\Rightarrow \frac{1}{k} \frac{dx}{dt} = \tanh\left(\frac{gt}{k}\right)$$

$$\Rightarrow dx = k \tanh\left(\frac{gt}{k}\right) dt \quad \dots \text{V-s-form}$$

on integration

$$\int dx = k \int \tanh\left(\frac{gt}{k}\right) dt + C_1$$

$$x = k \cdot \frac{\log \cosh\left(\frac{gt}{k}\right)}{\left(\frac{g}{k}\right)} + C_1$$

$$\boxed{x = \frac{k^2}{g} \log \cosh\left(\frac{gt}{k}\right) + C_1} \quad \text{--- (2)}$$

By initial conditions when $t=0$, $x=0$

$$\therefore 0 = \frac{k^2}{g} \log \cosh(0) + C_1$$

$$0 = 0 + C_1$$

$$\boxed{C_1 = 0}$$

eqⁿ (2) becomes

$$x = \frac{k^2}{g} \log \cosh \left(\frac{gt}{k} \right)$$

Hence proved.

⑤ One Dimensional Conduction of Heat (Heat Flow)

* Fourier law of Heat Conduction:-

"The rate of Heat conduction per unit area is directly proportional to the temperature gradient."

i.e., $\frac{q}{A} \propto \frac{dT}{dx}$

$$\frac{q}{A} = -K \frac{dT}{dx}$$

$$q = -KA \frac{dT}{dx}$$

where q = rate of heat transfer

K = proportionality constant
(Thermal conductivity)

A = Area of heat transfer

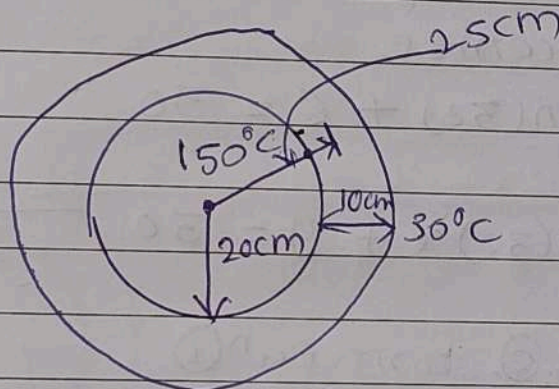
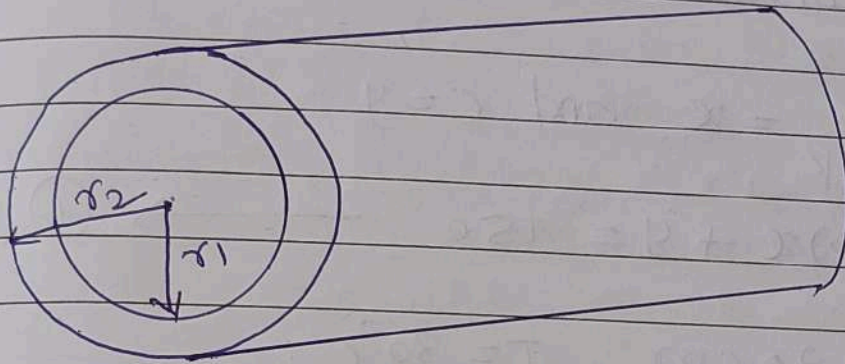
$\frac{dT}{dx}$ = temperature gradient

Ques) A steam pipe 40 cm in diameter contains steam at 150°C and is protected with a covering 10 cm thick for which $K = 0.0012$. If the temperature of outer surface of the covering is 30°C , find the temperature at a distance of 25 cm from the center of pipe under steady-state conditions.

⇒ Let q cal/sec be the constant quantity of heat flowing out radially through a surface of a pipe having radius r and length l cm.

Given $r = 20$ cm, $T = 150^\circ\text{C}$.

$r = 30 \text{ cm}$ then $T = 30^\circ \text{C}$
 $r = 25 \text{ cm}$ then $T = 9$



By Fourier law of heat conduction,

$$q = -KA \frac{dT}{dr}$$

but $A = 2\pi r$

$$q = -k(2\pi r) \frac{dT}{dr}$$

$$\therefore \frac{q}{-k2\pi r} dr = dT \quad \dots \text{V.S form}$$

on integration

$$\frac{-q}{2\pi k} \int \frac{1}{r} dr + C = \int dT$$

$$\therefore \left[\frac{-q}{2\pi k} \ln(r) + C = T \right] \longrightarrow \textcircled{A}$$

find C and $\frac{-q}{2\pi k}$.

As $r = 20 \text{ cm}$ then $T = 150^\circ \text{C}$

$\therefore$ eqn (A) becomes

$$\frac{-q}{2\pi K} \ln(20) + C = 150$$

put $\frac{-q}{2\pi K} = x$ and $C = y$

$$\ln(20)x + y = 150 \longrightarrow (1)$$

when $r = 30 \text{ cm}$, $T = 30^\circ \text{C}$.

$\therefore$ eqn (A) becomes

$$\frac{-q}{2\pi K} \ln(30) + C = 30$$

$$\ln(30)x + y = 30 \longrightarrow (2)$$

subtract eqn (2) from eqn (1)

$$\therefore \ln(20)x + y = 150$$

$$- \ln(30)x + y = 30$$

$$[\ln(20) - \ln(30)]x = 120$$

$$x = \frac{120}{\ln\left(\frac{2}{3}\right)}$$

$$x = -295.956 \Rightarrow \frac{-q}{2\pi K} = -295.956$$

put this value in eqn (2)

$$\ln(30)x(-295.956) + y = 30$$

$$y = 1036.606$$

substitute these values in eqn (A)

$$\therefore -295.956 \ln(r) + 1036.606 = T \longrightarrow (3)$$

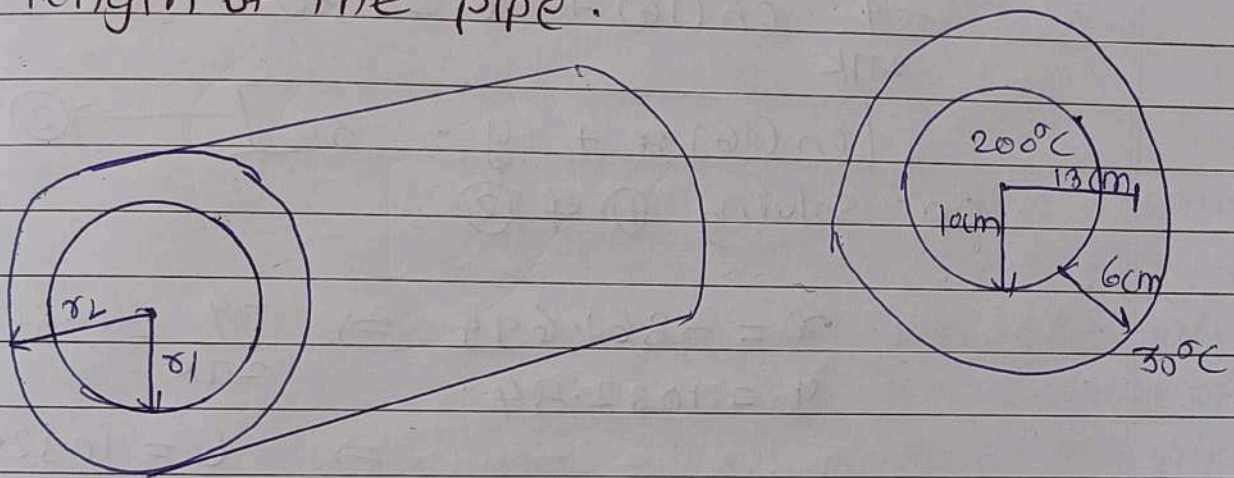
when $r = 25 \text{ cm}$

$$-295.956 \ln(25) + 1036.606 = T$$

$$83.96 = T$$

$$T = 83.96^\circ \text{C}$$

Que: A steam pipe 20 cm in diameter is protected with a covering 6 cm thick for which the coefficient of thermal conductivity is $k = 0.003$ cal/cm deg. sec in steady state. Find the temperature halfway through the covering, if the pipe surface is at 200°C and outer surface of the covering is at 30°C . Also find the heat lost per hour through a meter length of the pipe.



By Fourier law of Heat Conduction,

$$q = -KA \frac{dT}{dr}$$

but $A = 2\pi r$

$$q = -k(2\pi r) \frac{dT}{dr}$$

$$\therefore \frac{-q}{2\pi k r} dr = dT \quad \dots \text{V.S. Form}$$

on integration,

$$\frac{-q}{2\pi k} \int \frac{dr}{r} + C = \int dT$$

$$\frac{-q}{2\pi k} \ln(r) + C = T \quad \longrightarrow \textcircled{A}$$

When $r = 10\text{ cm}$, $T = 200^\circ\text{C}$.

$$\therefore \frac{-q}{2\pi K} \ln(10) + C = 200$$

put $x = \frac{-q}{2\pi K}$ and $y = C$

$$\boxed{\ln(10)x + y = 200} \longrightarrow \textcircled{1}$$

When $x = 16\text{cm}$, $T = 30^\circ\text{C}$

eqn (A) becomes

$$\frac{-q}{2\pi K} \ln(16) + C = 30$$

$$\boxed{\ln(16)x + y = 30} \longrightarrow \textcircled{2}$$

on solving (1) & (2)

$$x = -361.699 \Rightarrow \frac{-q}{2\pi K} = -361.699$$

$$y = 1032.84$$

$$\Rightarrow C = 1032.84$$

$\therefore$ eqn (A) becomes

$$\frac{-q}{2\pi K} \ln(x) + C = T$$

$$-361.699 \ln(x) + 1032.84 = T$$

When $x = 13\text{cm}$

$$-361.699 \ln(13) + 1032.84 = T$$

$$\boxed{T = 105.10^\circ\text{C}}$$

$\therefore$ As $\frac{-q}{2\pi K} = -361.699$

$$q = (2\pi K) \times 361.699$$

$$q = 2 \times 3.14 \times 0.003 \times 361.699$$

$$\boxed{q = 6.8144 \text{ cal/sec}}$$

$$1 \text{ hr} = 3600 \text{ sec.}$$

$$q = 6.8144 \times 3600 \text{ sec}$$

$$q = 24531.84 \text{ cal/hr}$$

This is heat lost for 1cm length of the pipe

$$1 \text{ m} = 100 \text{ cm}$$

$$q = 24531.84 \times 100$$

$$q = 2453184 \text{ cal/hr for}$$

a meter length of pipe

Que) H.W. A pipe 10cm in diameter contains steam at 100°C . It is covered with asbestos, 5cm thick for which $k = 0.0006$ and the outside surface is at 30°C . Find the amount of heat lost per hour from a meter long pipe

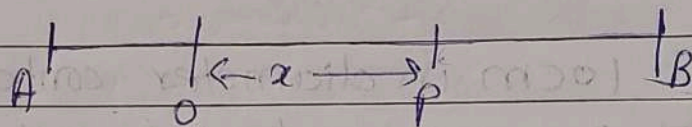
$$\Rightarrow q = 14000 \text{ cal/hr.}$$

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⑥ Simple Harmonic Motion: - (SHM)

A particle is said to execute S.H.M. if the particle moves in a straight line and the force acting on it is always directed towards a fixed point in the line and proportional to the distance of particle from the fixed point.

Let P be the position of particle at any time 't' moving in the line AB and let 'O' be the fixed point and of eq.



The eqn of motion of particle is

$$m\ddot{x} = F$$

$$m \frac{d^2x}{dt^2} = -m\omega^2 x$$

$$\boxed{\frac{d^2x}{dt^2} = -\omega^2 x} \longrightarrow (1)$$

on solving we get

$$V = \omega \sqrt{a^2 - x^2}$$

and

$$x = a \cos(\omega t)$$

$$x = a \cos \left[\omega \left(t + \frac{2\pi}{\omega} \right) \right]$$

where $\frac{2\pi}{\omega}$ is period of oscillations

* Frequency: The number of complete oscillations per second is called as the frequency.

$$n = \frac{1}{T} = \frac{\omega}{2\pi}$$

* Note: ① Maximum displacement on either side of 'o' is called as amplitude = a .

② frequency (n) = $\frac{\omega}{2\pi}$

③ We have maximum acceleration when velocity $v=0$.

Que) A particle executes S.H.M. When it is 2 cm from mid path, its velocity is 10 cm/sec and when it is 6 cm from centre of its path its velocity is 2 cm/sec. Find its period and its greatest acceleration.

⇒ The equation of S.H.M is

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\therefore v \frac{dv}{dx} = -\omega^2 x$$

$$v dv = -\omega^2 x dx \quad \dots \text{v.s. form}$$

on integration

$$\int v dv = -\omega^2 \int x dx + C_1$$

$$\frac{v^2}{2} = -\frac{\omega^2 x^2}{2} + C_1$$

$$v^2 = -\omega^2 x^2 + 2C_1$$

$$\boxed{v^2 = -\omega^2 x^2 + C} \quad \left(\because C = 2C_1 \right) \rightarrow \textcircled{1}$$

When $x = 2 \text{ cm}$, $v = 10 \text{ cm/sec}$.

∴ above eqn becomes

$$10^2 = -\omega^2(2)^2 + C$$

$$\boxed{100 = -4\omega^2 + C} \longrightarrow \textcircled{2}$$

when $x = 6 \text{ cm}$, $v = 2 \text{ cm/sec}$

eqn ① becomes

$$2^2 = -\omega^2(6)^2 + C$$

$$\boxed{4 = -36\omega^2 + C} \longrightarrow \textcircled{3}$$

Subtract eqn ② and eqn ③

$$100 = -4\omega^2 + C$$

$$-4 = -36\omega^2 + C$$

$$96 = 32\omega^2 + 0$$

$$\omega^2 = \frac{96}{32} = 3$$

$$\boxed{\omega^2 = 3} \Rightarrow \omega = \sqrt{3}$$

put this value in eqn ③

$$4 = -36(3) + C$$

$$4 + 108 = C$$

$$\boxed{C = 112}$$

$$\text{period} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{3}}$$

put $\omega^2 = 3$ and $C = 112$ in eqn ①.

$$\boxed{v^2 = -3x^2 + 112} \longrightarrow \textcircled{4}$$

∴ The greatest acceleration is achieved when $x = a$ and $v = 0$

$$\therefore 0^2 = -3a^2 + 112$$

$$3a^2 = 112$$

$$a^2 = \frac{112}{3}$$

$$a = \sqrt{\frac{112}{3}}$$

$$\text{Acceleration} = \frac{d^2x}{dt^2} = -\omega^2 x$$

When $x=a \Rightarrow$ greatest accelⁿ.

$$\left(\frac{d^2x}{dt^2} \right)_{x=a} = -\omega^2 a$$

$$\left(\frac{d^2x}{dt^2} \right)_{x=a} = -3\sqrt{\frac{112}{3}}$$

$$\left(\frac{d^2x}{dt^2} \right)_{x=a} = -\sqrt{336}$$

$\therefore$ Greatest acceleration is $\sqrt{336}$ directed towards the origin.

Que A particle moving in a straight line with SHM has velocity 15 cm/sec and 12 cm/sec respectively. When it is distance is at 8 cm and 4 cm from the centre of its path. Find the period and amplitude.

H.W.

$$\Rightarrow \omega^2 = \frac{81}{7}, \quad C = \frac{2304}{7}, \quad T = 1.84 \text{ sec.}$$

$$a = \frac{16}{3} = 5.33 \text{ cm.}$$